

Generated-Algebra Correlations Reconstruct Tensor Factors Only After Split or Kernel Separation

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Abstract

Split inclusions and W^* -tensor-product independence are standard positive hypotheses in algebraic quantum field theory: when the multiplication map $\mu : \mathcal{A} \bar{\otimes} \mathcal{B} \rightarrow \mathcal{A} \vee \mathcal{B}$ is a normal isomorphism, the generated algebra determines the spatial tensor-product coordinate. This note isolates the converse boundary. If the supplied data are only generated-algebra correlations $\gamma(\mu(x))$, then a candidate tensor-product element $x \in \mathcal{A} \bar{\otimes} \mathcal{B}$ is reconstructed exactly modulo the common tested multiplication kernel. In particular, a shared nontrivial central projection $p \in \mathcal{A} \cap \mathcal{B}$ gives $0 \neq p \otimes (1 - p) \in \ker \mu$. Thus generated-algebra tomography is not, by itself, a split/tensor-factor theorem. The positive exit is exact: once split or W^* -tensor-product independence makes μ a normal isomorphism, μ^{-1} recovers the unquotiented tensor coordinate.

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1 Primary sources

The boundary below was checked against Doplicher–Longo split inclusions [1], Buchholz–Wichmann nuclearity and causal independence [2], and Werner local preparability [3]. These works are used here as positive routes to split independence, not as claims that split theory is absent from physics.

2 Generated algebra versus spatial tensor product

Let $\mathcal{A}, \mathcal{B} \subset B(\mathcal{H})$ be commuting von Neumann algebras, and let $\mathcal{A} \vee \mathcal{B}$ be the von Neumann algebra they generate. Suppose multiplication extends to a normal representation

$$\mu : \mathcal{A} \bar{\otimes} \mathcal{B} \longrightarrow \mathcal{A} \vee \mathcal{B}, \quad \mu(a \otimes b) = ab.$$

Let $\Gamma \subset (\mathcal{A} \vee \mathcal{B})_*$ be the normal functionals used as supplied generated-algebra correlations:

$$D_\Gamma(x) = (\gamma(\mu(x)))_{\gamma \in \Gamma}, \quad x \in \mathcal{A} \bar{\otimes} \mathcal{B}.$$

Theorem 1 (multiplication-kernel criterion). *The data D_Γ reconstruct the spatial tensor-product element x if and only if*

$$\bigcap_{\gamma \in \Gamma} \ker(\gamma \circ \mu) = \{0\}. \quad (1)$$

If Γ separates points of $\mathcal{A} \vee \mathcal{B}$, then (1) is equivalent to

$$\ker \mu = \{0\}. \quad (2)$$

Thus full generated-algebra tomography reconstructs the unquotiented spatial tensor-product coordinate exactly when the multiplication representation is injective.

Proof. Two candidates $x, y \in \mathcal{A} \bar{\otimes} \mathcal{B}$ have the same supplied data exactly when

$$\gamma(\mu(x - y)) = 0 \quad (\gamma \in \Gamma),$$

that is, when $x - y$ lies in the common kernel in (1). Therefore the data separate all candidates precisely when that kernel is zero. If Γ separates $\mathcal{A} \vee \mathcal{B}$, then $\gamma(\mu(z)) = 0$ for all $\gamma \in \Gamma$ is equivalent to $\mu(z) = 0$, giving (2). $\square$

Proposition 1 (shared-center witness). *If $\mathcal{A} \cap \mathcal{B}$ contains a projection $0 \neq p \neq 1$, then*

$$0 \neq p \otimes (1 - p) \in \mathcal{A} \bar{\otimes} \mathcal{B}, \quad \mu(p \otimes (1 - p)) = 0.$$

Consequently no data package that factors only through $\mathcal{A} \vee \mathcal{B}$ can distinguish the zero tensor-product element from this nonzero tensor-kernel direction.

Proof. The projections p and $1 - p$ are nonzero in the two tensor factors, so $p \otimes (1 - p)$ is a nonzero positive element of the spatial tensor product. Because p is a common projection, multiplication sends it to $p(1 - p) = 0$. Any datum that factors through μ assigns the same value to this element and to 0. $\square$

Proposition 2 (split inverse recovers the tensor coordinate). *Assume the target AQFT hypotheses supply a split inclusion, type-I interpolation, or W^* -tensor-product independence strong enough that*

$$\mu : \mathcal{A} \bar{\otimes} \mathcal{B} \longrightarrow \mathcal{A} \vee \mathcal{B}$$

is a normal isomorphism. Then full generated-algebra data determine the unquotiented spatial tensor-product coordinate: for every $y \in \mathcal{A} \vee \mathcal{B}$, the unique tensor coordinate is $\mu^{-1}(y)$. Equivalently, for every $x \in \mathcal{A} \bar{\otimes} \mathcal{B}$, the generated-algebra element $\mu(x)$ determines x .

Proof. A normal isomorphism is injective and has a normal inverse onto its range, which is $\mathcal{A} \vee \mathcal{B}$. Hence if $\mu(x) = \mu(x')$, then applying μ^{-1} gives $x = x'$. Conversely, every generated-algebra element $y \in \mathcal{A} \vee \mathcal{B}$ has exactly one preimage under μ , namely $\mu^{-1}(y)$. The kernel obstruction in Theorem 1 is therefore absent. $\square$

3 Positive exits

Exit premise	Effect
Split inclusion or type-I interpolation	Supplies the standard W^* -tensor-product independence structure for the bipartition; when this makes μ a normal isomorphism, Proposition 2 applies.
Normal isomorphism $\mathcal{A} \bar{\otimes} \mathcal{B} \simeq \mathcal{A} \vee \mathcal{B}$	Makes μ injective; the inverse of μ reconstructs the tensor coordinate.
Kernel-separating tests	Adds data that distinguish directions in $\ker \mu$.
Quotient, factor, or inadmissibility axiom	Removes the kernel directions from the physical target instead of reconstructing them.

The exact external question is:

Does the target AQFT setup assume split/ W^* -tensor-product independence, type-I interpolation, factor-valuedness, or kernel-separating tests for the bipartition? If not, generated-algebra correlations reconstruct only the quotient of the candidate spatial tensor product by $\ker \mu$.

If the split hypothesis is supplied, it is a positive exit from the Team 1 finite-center obstruction. If it is absent, the Team 1 row is only the standard multiplication-kernel boundary, not a new split theorem.

References

- [1] S. Doplicher and R. Longo, *Standard and split inclusions of von Neumann algebras*, Invent. Math. 75 (1984), 493–536, doi:10.1007/BF01388641.
- [2] D. Buchholz and E. H. Wichmann, *Causal independence and the energy-level density of states in local quantum field theory*, Commun. Math. Phys. 106 (1986), 321–344, doi:10.1007/BF01454978.
- [3] R. F. Werner, *Local preparability of states and the split property in quantum field theory*, Lett. Math. Phys. 13 (1987), 325–329, doi:10.1007/BF00401161.